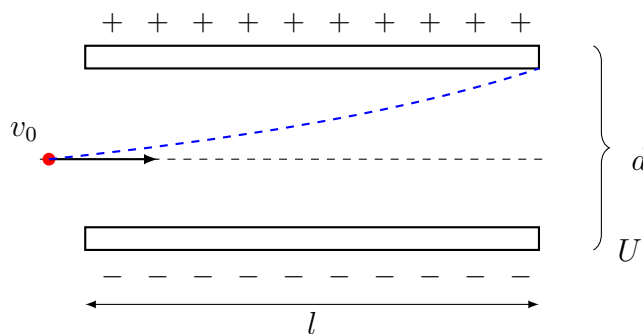


(KTXFI1EBNF) Physics I. Test 1

May 11, 2026

- At what distance from Pluto does a Pluto-stationary satellite orbit in the plane of the equator if it remains constantly above the same point on Pluto's surface? ($M_{\text{Pluto}} = 1.305 \cdot 10^{22} \text{ kg}$, $T_{\text{rot}} = 6.38723 \text{ day}$, $R_{\text{Pluto}} = 1188 \text{ km}$, $G = 6.6743 \cdot 10^{-11} \text{ Nm}^2/\text{kg}^2$.) (10 points)
- A 5 kg object slides down a 4-meter-high incline that makes an angle of 45° with the horizontal. What is the speed of the object when it reaches the bottom of the incline if it starts from rest at the top and:
 - there is no friction between the object and the incline?
 - the coefficient of friction is 0.1?
 ($g = 9.81 \text{ m/s}^2$) (10 points)
- At a temperature of 30°C , a steel shaft with a diameter of 11.30 cm needs to be fitted with an aluminum ring with an inner diameter of 11.27 cm at the same temperature. The linear thermal expansion coefficient of steel is $12 \cdot 10^{-6} \text{ 1/K}$, and the linear thermal expansion coefficient of aluminum is $28.7 \cdot 10^{-6} \text{ 1/K}$. What temperature must the ring be heated to? (10 points)
- A weather balloon floating at an altitude of several kilometers has a volume of 1800 m^3 . The temperature and pressure of the helium inside it are the same as those of the surroundings: -50°C and $7 \cdot 10^3 \text{ Pa}$, respectively.
 - How many moles of helium are in the balloon?
 - What is the mass of the helium gas? ($M = 4 \frac{\text{g}}{\text{mol}}$)
 - What was the volume of the balloon when it was launched from Earth, if the gas inside the balloon was characterized by a temperature of 0°C and a pressure of 10^5 Pa ?
 ($R = 8.31 \text{ J/K}$) (10 points)
- An electron enters the region between the parallel plates shown in the figure halfway between the plates. The length of the plates is $l = 12 \text{ cm}$, their separation is $d = 2 \text{ cm}$, and the voltage between them is $U = 300 \text{ V}$. What must the initial velocity of the electron be so that it just exits the region between the plates?



$$(m_e = 9.1 \cdot 10^{-31} \text{ kg}, e = 1.6 \cdot 10^{-19} \text{ C}) \quad (10 \text{ points})$$

Requirement for obtaining the course signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).